

Dark Matter in the Universe: Evidence, Candidates, and Detection Strategies

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Abstract

Dark matter is a crucial component of the universe, playing a fundamental role in the formation and evolution of cosmic structures, yet its physical nature remains unknown. This review introduces the concept of dark matter by outlining the historical and observational motivations for its existence, emphasizing discrepancies between visible matter and gravitational effects at galactic and cosmological scales [1,2]. The core content discusses key observational evidence such as galaxy rotation curves, gravitational lensing, and measurements of the cosmic microwave background, which collectively support the presence of a non-luminous matter component [1,14]. The review further examines major theoretical proposals for dark matter candidates [6,7] and summarizes current experimental approaches aimed at detecting dark matter through astrophysical observations, underground experiments, and high-energy particle collisions[8–10]. In the final part, the review highlights unresolved challenges, including the lack of direct detection and tensions in small-scale structure formation, while emphasizing future observational missions and experimental advancements[4,13]. The novelty of this review lies in presenting a coherent and conceptually accessible synthesis of observational evidence and detection efforts, seeking a comprehensive understanding of the dark matter problem in modern astrophysics.

Keywords

Dark matter, galaxy rotation curves, gravitational lensing, cosmic microwave background, large-scale structure, dark matter candidates, detection methods, cosmology, astrophysics.

1. Introduction

Understanding the composition of the universe has been one of the central goals of modern astrophysics. Observations across a wide range of spatial scales indicate that the matter directly visible through electromagnetic radiation constitutes only a small fraction of the total matter content of the universe [1]. A significant discrepancy exists between the observed luminous matter and the gravitational effects required to explain the dynamics of galaxies, galaxy clusters, and the large-scale structure of the universe [2,15]. This discrepancy has led to the introduction of the concept of dark matter, a form of matter that interacts primarily through gravity and remains undetectable through conventional electromagnetic observations.

The existence of dark matter was first suggested through studies of galactic motion and has since been supported by multiple independent observational probes. These include the flat rotation curves of spiral galaxies, gravitational lensing effects in massive structures, and precise measurements of temperature fluctuations in the cosmic microwave background [1,14]. Together, these observations provide compelling evidence that dark matter is a dominant component shaping cosmic evolution.

Despite its well-established gravitational influence, the fundamental nature of dark matter remains unknown. Identifying its properties and origin represents one of the most profound challenges in contemporary astrophysics, motivating extensive theoretical modeling and experimental efforts worldwide [6,15].

2. Observational Evidence for Dark Matter

The hypothesis of dark matter is not based on a single observation but emerges from a wide range of independent astrophysical measurements spanning galactic, extragalactic, and cosmological scales [2,14]. In each case, the gravitational behavior of astrophysical systems cannot be explained solely by the presence of visible matter such as stars, gas, and dust [1]. Instead, an additional, non-luminous component is required to account for the observed dynamics. This section reviews the most significant observational evidence supporting the existence of dark matter.

2.1 Galaxy Rotation Curve

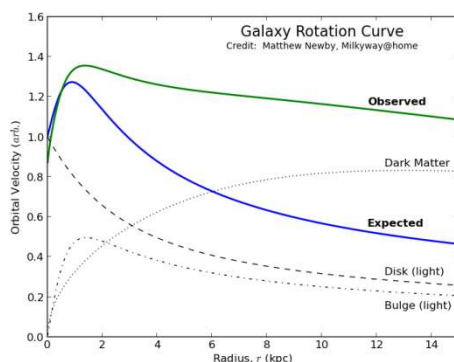


Figure 1 — Galaxy Rotation Curve (Observed vs Expected Models)

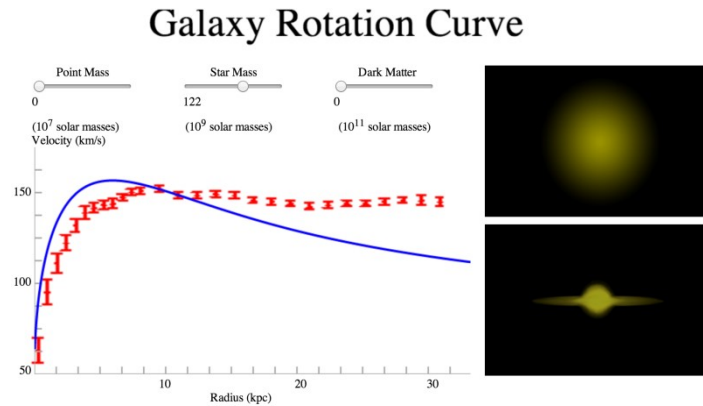


Figure 2— Galaxy Rotation Curve Comparison Across Mass Components

One of the earliest and most compelling pieces of evidence for dark matter arises from the study of rotation curves of spiral galaxies. According to classical gravitational theory, the orbital velocity of stars and gas in a galaxy is expected to decrease with increasing distance from the galactic center once most of the luminous mass has been enclosed. However, observations reveal that the rotational velocities of stars in spiral galaxies remain approximately constant well beyond the visible edge of the stellar disk [2,15].

This phenomenon, known as flat rotation curves, implies that the enclosed mass of galaxies continues to increase with radius, even in regions where little or no luminous matter is detected [2]. The discrepancy between the expected and observed rotational behavior cannot be resolved by modifying the distribution of visible matter alone. Instead, it suggests the presence of an extended, massive halo composed of unseen matter surrounding the galaxy [15].

Detailed studies across a wide range of galaxy types, including dwarf galaxies and low surface brightness galaxies, consistently show similar behavior. These systems are particularly important because they contain relatively small amounts of luminous matter, making the effects of dark matter even more pronounced. The universality of flat rotation curves across galaxies of different sizes and morphologies strongly supports the idea that dark matter is a fundamental and ubiquitous component of galactic structure [4].

2.2 Gravitational Lensing



Figure 3— Strong Gravitational Lensing in Galaxy Cluster

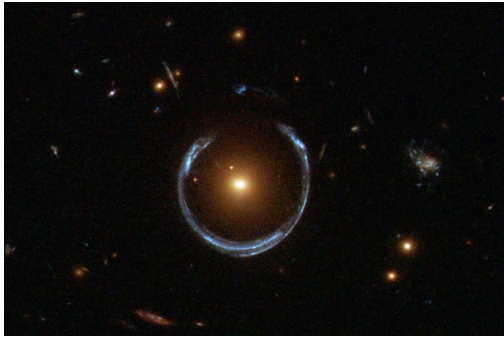


Figure 4— Einstein Ring Formation by Gravitational Lensing

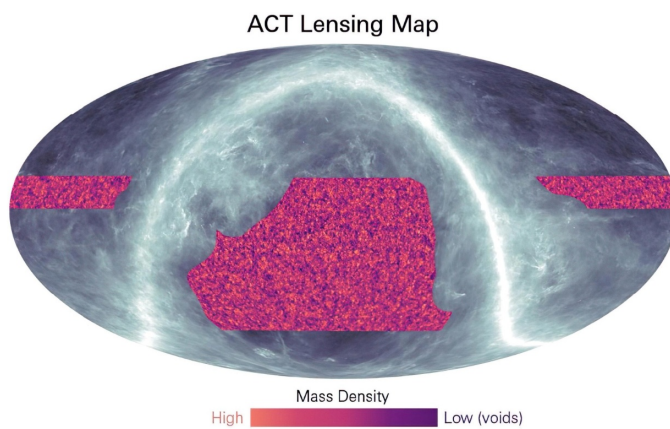


Figure 5— ACT Mass Density Lensing Map of the Sky

Gravitational lensing provides one of the most direct methods for probing the distribution of mass in the universe, independent of the nature or state of the matter involved [14]. According to the theory of gravity, massive objects curve the surrounding spacetime, causing light from background sources to bend as it passes near them. The amount of bending depends only on the total mass of the lensing object, regardless of whether that mass is luminous or dark [15].

Observations of gravitational lensing in galaxy clusters reveal mass distributions that far exceed what can be accounted for by visible matter alone. In many cases, lensing maps show that the majority of the mass is distributed in regions with little optical emission, indicating a dominant contribution from dark matter. Both strong lensing, which produces arcs and multiple images of background galaxies, and weak lensing, which results in subtle distortions of galaxy shapes, consistently point toward the presence of large amounts of unseen mass [14].

Particularly striking evidence comes from systems in which ordinary matter and dark matter appear spatially separated due to dynamic interactions. In such cases, the gravitational lensing signal traces the dominant mass component rather than the luminous gas, reinforcing the interpretation that dark matter behaves as a collisionless substance that interacts primarily through gravity [15].

2.3 Galaxy Clusters and Large-Scale Structures

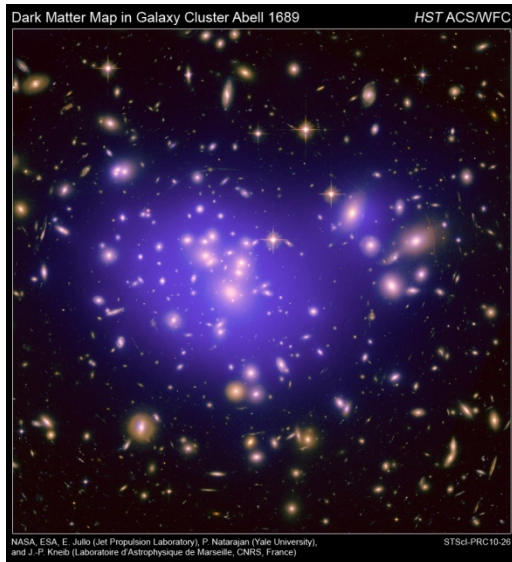


Figure 6— Dark Matter Distribution in Galaxy Cluster Abell 1689

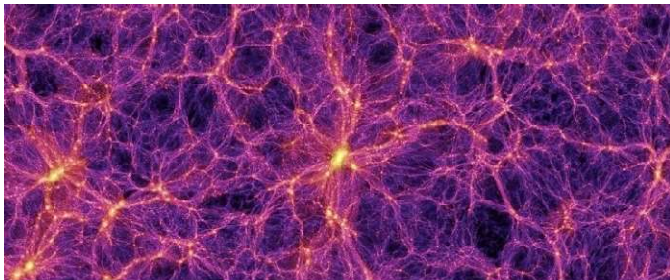


Figure 7— Cosmic Web Simulation Structure Map

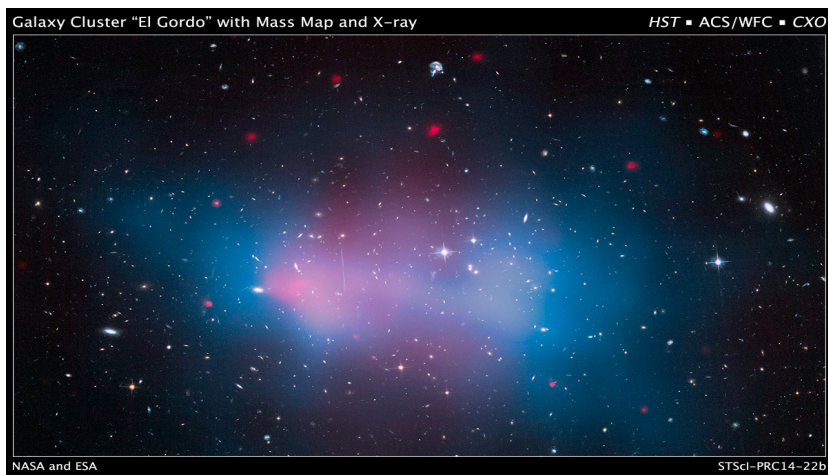


Figure 8— Galaxy Cluster "El Gordo" — Combined Mass & X-Ray Map

Galaxy clusters are the largest gravitationally bound structures in the universe and serve as powerful laboratories for testing mass distributions on large scales. Measurements of galaxy velocities within clusters indicate gravitational potentials that are far deeper than what luminous matter alone can generate. Independent mass estimates derived from X-ray observations[2,15] of hot intracluster gas and gravitational lensing consistently confirm the presence of a dominant dark matter component.

On even larger scales, the distribution of galaxies throughout the universe reveals a complex network of filaments, sheets, and voids known as the cosmic web. Numerical simulations of structure formation show that such large-scale patterns naturally arise when dark matter dominates the matter content of the universe [1,5]. In contrast, models that exclude dark matter fail to reproduce the observed distribution and evolution of cosmic structures [5].

2.4 Cosmic Microwave Background Constraints

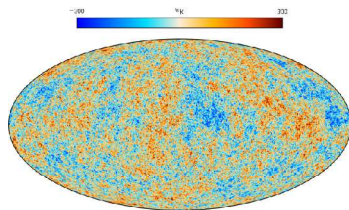


Figure 9— CMB Temperature Anisotropy Full-Sky Map

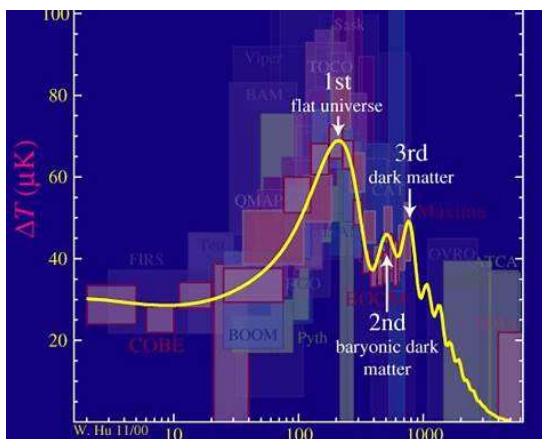


Figure 10— CMB Acoustic Peaks (Power Spectrum Graph)

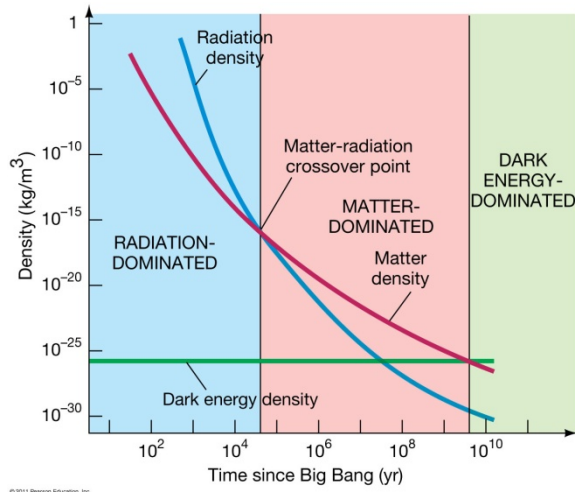


Figure 11— Cosmological Density Evolution Over Time

Precise measurements of temperature fluctuations in the cosmic microwave background provide strong constraints on the composition of the universe. The pattern of these fluctuations reflects conditions in the early universe and encodes information about the relative contributions of different forms of matter and energy.

The observed anisotropy spectrum can only be accurately explained if a substantial fraction of matter is non-luminous and weakly interacting. Dark matter plays a crucial role in enhancing early density fluctuations, allowing structures such as galaxies to form within the available cosmic time. Without dark matter, the observed level of structure in the universe would be difficult to reconcile with the initial conditions inferred from the cosmic microwave background.

3. Role of Dark Matter in Structure Formation

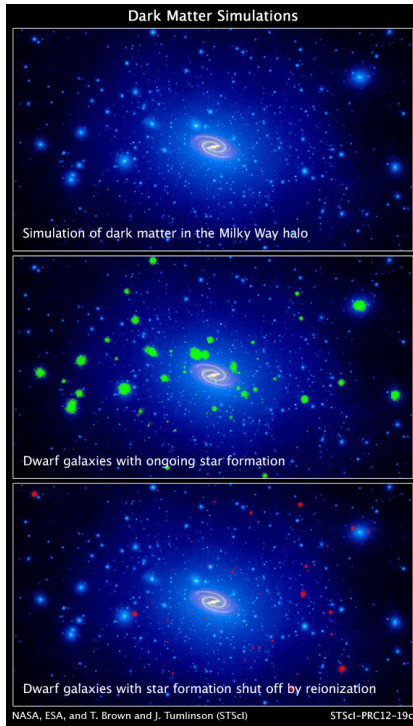


Figure 12— Dark Matter Halo Simulation (Milky Way & Dwarfs)

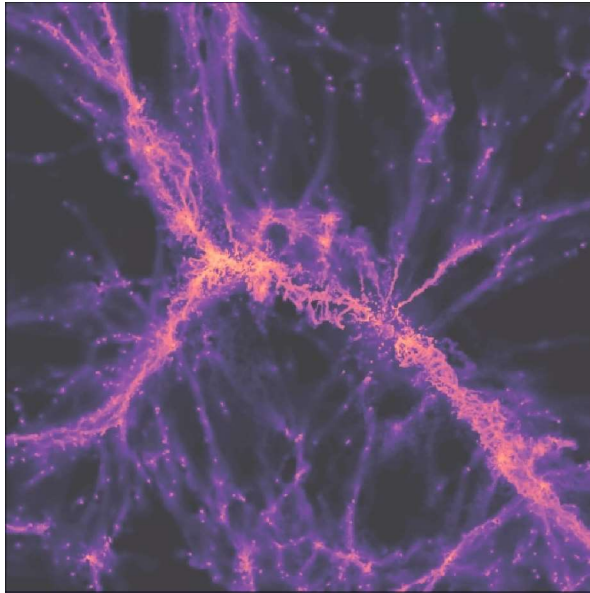


Figure 13— Cosmic Web Filament Simulation

The formation of cosmic structures such as galaxies, galaxy clusters, and the large-scale filamentary network of matter observed in the universe is strongly influenced by the presence of dark matter. In the early universe, small density fluctuations [2,3] existed in the primordial matter distribution. While ordinary matter [1] was tightly coupled to radiation and resisted gravitational collapse during the early epochs, dark matter, due to its weak interaction with

radiation, was able to respond to gravitational instabilities at much earlier times. This difference played a crucial role in shaping the subsequent evolution of cosmic structures.

Dark matter acts as the gravitational backbone of structure formation. Regions with slightly higher dark matter density gradually grew through gravitational attraction, forming potential wells into which ordinary matter later fell after the universe cooled sufficiently. As baryonic matter accumulated within these dark matter halos, it underwent further processes such as cooling, fragmentation, and star formation, eventually giving rise to galaxies. Without dark matter, the growth of structures would be significantly delayed, making it difficult to explain the existence of galaxies and clusters within the observed age of the universe.

Numerical simulations of cosmic evolution provide strong support for this framework. Simulations that include dark matter successfully reproduce the observed large-scale structure of the universe, including the cosmic web [3] composed of filaments, sheets, and voids. In contrast, models that exclude dark matter fail to generate the correct distribution and clustering of galaxies. These results demonstrate that dark matter dominates the gravitational dynamics on large scales and governs the hierarchical formation of structures, where smaller systems merge over time to form larger ones.

On smaller, galactic scales, dark matter halos determine the overall mass distribution and stability of galaxies. The properties of these halos influence galaxy rotation curves, star formation efficiency, and the survival of satellite galaxies. However, discrepancies between simulations and observations on small scales, such as the distribution of satellite galaxies and central density profiles, remain active areas of research. These challenges highlight the importance of refining both theoretical models and observational constraints while reinforcing the central role of dark matter in cosmic structure formation [4,13].

4. Theoretical Models and Dark Matter Candidates

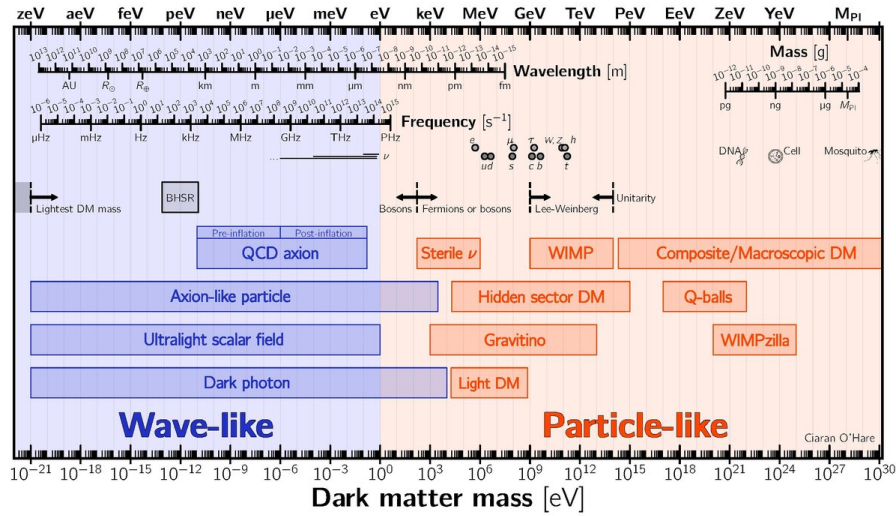


Figure 14— Dark Matter Candidates Mass–Wavelength Parameter Space

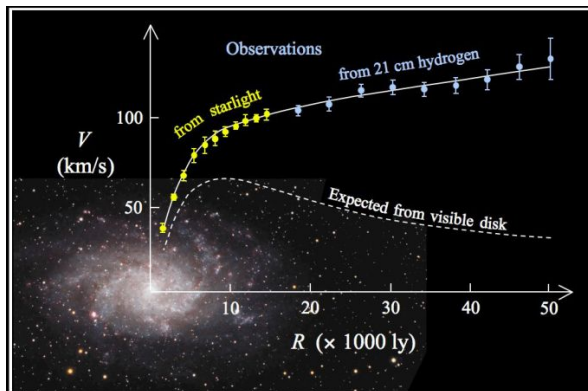


Figure 15— Predicted vs Observed Galactic Rotation via Dark Matter Mass

Although the gravitational effects of dark matter are well established through observations, its fundamental nature remains unknown. The inability of ordinary matter to account for the observed mass discrepancies has led to the development of several theoretical models proposing new forms of matter [6,15] beyond those directly detected through electromagnetic interactions. These models aim to explain the stability, abundance, and large-scale distribution of dark matter while remaining consistent with cosmological and astrophysical constraints.

A successful dark matter model must satisfy several key requirements. It must be stable on cosmological time scales, interact weakly or negligibly with electromagnetic radiation, and

possess the correct abundance to match observations of cosmic structure and background radiation. Additionally, it must be compatible with observations on both large and small astrophysical scales. These criteria have guided the development of various dark matter candidates, which can be broadly classified based on their physical properties and cosmological behavior [6,7].

4.1 Cold Dark Matter Framework

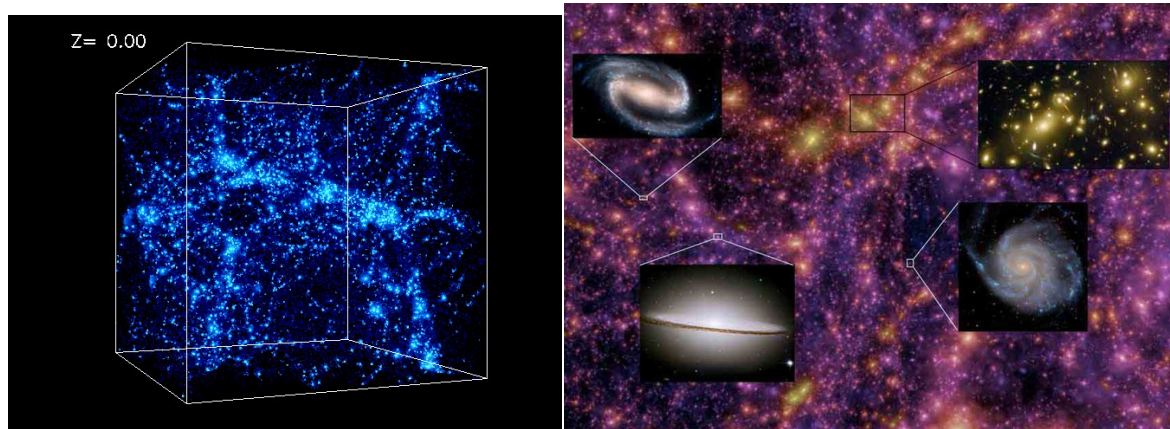


Figure 16— Cold Dark Matter Structure Formation Simulation

The cold dark matter framework is the most widely accepted theoretical model for explaining the formation of cosmic structures. In this framework, dark matter particles are assumed to move slowly compared to the speed of light during the early stages of the universe. This property allows small density fluctuations to grow efficiently under gravity, leading to the hierarchical formation of structures in which small objects form first and later merge to create larger systems such as galaxies and galaxy clusters.

Cold dark matter successfully explains the observed large-scale structure of the universe, including the filamentary cosmic web and the clustering properties of galaxies. Numerical simulations [2,3] based on this framework reproduce many observational features with remarkable accuracy. As a result, cold dark matter has become the standard reference model for studying cosmic evolution and structure formation.

However, on smaller, galactic scales, certain discrepancies remain. These include differences between predicted and observed density distributions in galaxy centers and the number of small satellite galaxies. While these challenges do not invalidate the cold dark matter framework [4], they suggest that additional physical processes or refinements may be necessary to fully reconcile theory with observation.

4.2 Particle-Based Dark Matter Candidates

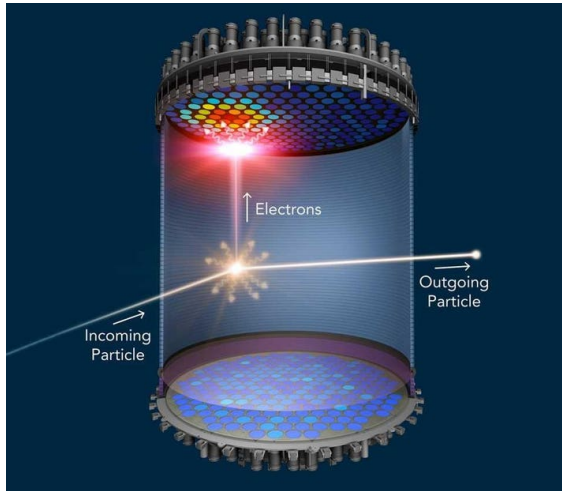


Figure 17— Particle Detection Chamber Visualization for WIMP Search

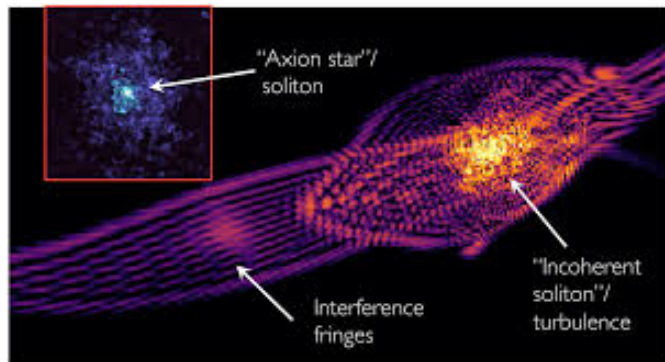


Figure 18— Axion Star/Soliton Dark Matter Visualization

Many dark matter models propose the existence of new elementary particles that interact weakly with ordinary matter. These particle-based candidates arise naturally in extensions of known physical theories and offer testable predictions for experimental searches. Such particles are typically electrically neutral, massive, and stable, allowing them to remain undetected through conventional observational techniques while exerting gravitational influence.

One class of proposed candidates consists of heavy particles that interact through forces comparable in strength to the weak nuclear interaction. These particles naturally produce the correct dark matter abundance through processes occurring in the early universe. Another class involves extremely light particles that behave as coherent fields on astrophysical scales. Despite their very small masses, these particles can contribute significantly to the total matter density due to their large number.

Sterile neutrino-like candidates represent another possibility, characterized by their weak interactions and intermediate mass range. These candidates can influence the formation of structures on small scales and may help address some of the discrepancies observed in galaxy distributions. Each of these particle-based models is subject to ongoing theoretical refinement and experimental testing [6,7].

4.3 Warm and Alternative Dark Matter Models

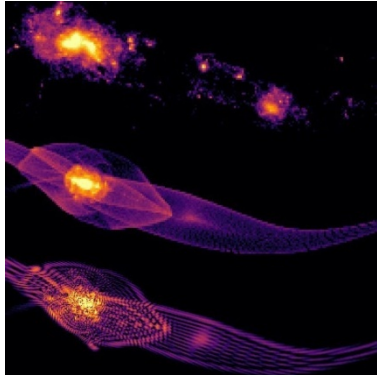


Figure 19— Wave-like (Axion/Fuzzy) Dark Matter Simulation Showing Solitonic Cores and Interference Patterns

In addition to cold dark matter, alternative models have been proposed in which dark matter particles possess moderate thermal motion in the early universe. These warm dark matter models suppress the formation of very small-scale structures while preserving large-scale features. As a result, they offer potential explanations for some of the small-scale discrepancies associated with cold dark matter predictions.

Other alternative approaches explore modifications to the interaction properties or self-interactions of dark matter. In such models, dark matter particles may interact with one another through non-gravitational forces, altering the internal structure of dark matter halos. These interactions can lead to smoother density distributions in galactic centers and affect the dynamics of galaxy clusters.

While alternative models remain less established than the cold dark matter framework, they provide valuable insights into the range of physical possibilities and highlight the importance of confronting theoretical predictions with increasingly precise observational data [13].

4.4 Model Constraints and Theoretical Challenges

Theoretical models of dark matter are constrained by a wide array of astrophysical and cosmological observations. Measurements of large-scale structure, gravitational lensing, and background radiation place strong limits on the allowed properties of dark matter candidates.

Any viable model must simultaneously explain these observations without conflicting with established physical principles.

One of the primary challenges lies in the absence of direct experimental confirmation of any proposed dark matter particle. Despite extensive efforts, no definitive non-gravitational interaction has yet been observed. This has motivated the exploration of broader theoretical frameworks and the development of more sensitive experimental techniques.

The diversity of dark matter models reflects both the complexity of the problem and the richness of possible solutions. Continued progress will require close collaboration between theory, observation, and experimentation, with future discoveries expected to significantly refine or reshape current understanding [15].

5. Detection Methods of Dark Matter



Figure 20— Direct Detection Experimental Setup (Underground Laboratory)

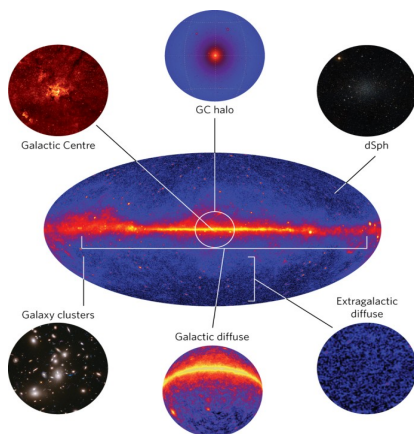


Figure 21— Astrophysical Indirect Detection Targets (Gamma-ray & Neutrino Sources)

Despite the strong gravitational evidence supporting the existence of dark matter, detecting it through non-gravitational interactions remains one of the most challenging problems in modern astrophysics and particle physics. Since dark matter does not emit or absorb electromagnetic radiation, its detection relies on indirect signatures or extremely weak interactions with ordinary matter. Over the past several decades, a wide range of experimental strategies have been developed to identify the physical nature of dark matter. These approaches can be broadly categorized into direct detection, indirect detection, and collider-based searches [8-10].

5.1 Direct Detection Methods

Direct detection experiments aim to observe the interaction of dark matter particles with ordinary matter in controlled laboratory environments. These experiments are typically conducted deep underground to shield sensitive detectors from cosmic radiation and other background noise. The basic principle involves measuring the tiny amount of energy deposited when a dark matter particle scatters off an atomic nucleus within the detector material.

A variety of detector technologies are employed, including cryogenic solid-state detectors, liquid noble gas detectors, and scintillation-based systems. These detectors are designed to measure recoil energies that are extremely small, often requiring advanced noise reduction and background discrimination techniques. The expected interaction rates are low, making long exposure times and large detector masses essential for improving sensitivity.

Although no confirmed detection has yet been achieved, direct detection experiments have significantly improved their sensitivity over time. Current results place strong constraints on the interaction strength between dark matter and ordinary matter, ruling out large regions of parameter space for several theoretical models. The continued development of more massive detectors with lower background levels is expected to further enhance the discovery potential of this approach [8,9].

5.2 Indirect Detection through Astrophysical Signals

Indirect detection methods search for secondary particles produced when dark matter particles interact with one another or decay in astrophysical environments. These interactions may generate observable signals such as high-energy photons, neutrinos, or charged particles. Regions with high dark matter density, including the centers of galaxies, galaxy clusters, and dwarf satellite galaxies, are prime targets for indirect searches.

High-energy gamma rays are among the most widely studied indirect signatures. Observations of excess gamma radiation from specific regions of the sky can potentially be linked to dark matter interactions. Similarly, neutrino observatories search for neutrinos originating from dark matter accumulation and interactions within massive celestial bodies such as the Sun or Earth.

One of the main challenges of indirect detection lies in distinguishing potential dark matter signals from conventional astrophysical sources. Many high-energy processes, including supernova remnants and active galactic nuclei, can produce similar observational signatures. As a result, indirect detection requires careful modeling of astrophysical backgrounds and often relies on correlations across multiple observational channels to strengthen interpretations [10,12].

5.3 Collider-Based Searches

Collider-based searches attempt to produce dark matter particles in high-energy particle collisions. In these experiments, dark matter particles are not detected directly but are inferred through missing energy and momentum in the final states of particle interactions. If dark matter particles are created during a collision, they would escape the detector without interacting, leading to an apparent imbalance in the observed energy.

These searches provide a complementary approach to astrophysical detection methods, as they probe dark matter interactions at energy scales inaccessible to direct and indirect experiments. Collider experiments are particularly valuable for testing theoretical models that predict dark matter particles with specific interaction properties.

However, interpreting collider results requires careful theoretical modeling, as missing energy signatures can also arise from other known processes. While no definitive evidence for dark matter production has yet been observed in collider experiments, the resulting constraints play an important role in narrowing down viable theoretical models [6,11].

5.4 Complementarity of Detection Approaches

Each detection method probes different aspects of dark matter properties and is subject to distinct experimental limitations. Direct detection focuses on local dark matter interactions, indirect detection explores astrophysical environments with high dark matter densities, and collider searches investigate particle interactions at high energies. Together, these approaches form a complementary framework for understanding dark matter.

The absence of a confirmed signal across all detection strategies highlights the complexity of the dark matter problem. Rather than indicating failure, this situation has driven the development of more sensitive experiments and encouraged broader theoretical exploration. Future progress is expected to arise from the combined analysis of results across multiple detection methods, supported by advances in observational capabilities and theoretical modeling [15].

6. Conclusion and Novelty

Dark matter remains one of the most profound unsolved problems in modern astrophysics, with its existence firmly supported by a wide range of observational and cosmological evidence. From galaxy rotation curves and gravitational lensing to large-scale structure formation and measurements of the cosmic microwave background, observations consistently indicate the presence of a dominant non-luminous matter component shaping the universe. Despite decades of research, the fundamental nature of dark matter continues to elude direct detection, motivating sustained theoretical and experimental efforts across multiple disciplines.

This review has presented a coherent synthesis of recent developments in dark matter research, emphasizing observational foundations, its role in cosmic structure formation, contemporary theoretical models, and current detection strategies. By focusing on results and interpretations discussed in recent literature, the review highlights how advances in simulations, precision cosmology, and experimental sensitivity have significantly refined constraints on dark matter properties while also exposing persistent challenges, particularly on small astrophysical scales and in direct detection experiments.

The novelty of this review lies in its integrated and conceptually accessible presentation of post-2019 developments, bridging observational astrophysics, theoretical modeling, and experimental approaches within a unified framework. Rather than concentrating on a single detection strategy or model class, the review emphasizes the complementarity of methods and the evolving nature of constraints derived from recent studies. This approach provides early-stage researchers with a clear and up-to-date understanding of the dark matter problem, while also identifying open questions that are likely to guide future research in astrophysics and cosmology.

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